

Amazon Aurora Global Database Setup

To Begin with the Lab

Summary of the Lab

In this lab, an existing **Amazon Aurora** cluster in the Ohio region was converted into an **Aurora Global Database** by adding another AWS Region. The primary cluster already existed, and the new region was added from the cluster actions menu to create a second cluster in a different region. The video explained that the secondary region requires its own VPC and database subnet group, but most of the cluster configuration remains the same. The secondary cluster can also use its own Aurora Capacity Units if it is a serverless setup. Finally, the lab emphasized that changes made in the primary region are synchronized to the secondary region.

Understanding Amazon Aurora Global Database

An **Amazon Aurora Global Database** is a multi-region setup where one Aurora cluster acts as the primary and one or more clusters in other regions act as secondary replicas. It exists so applications can serve users in different regions with lower latency and have disaster recovery options if the primary region becomes unavailable. The secondary region is not just a copy of settings; it is a real Aurora cluster in another AWS Region with its own networking and database subnet group. This makes the setup useful when you need global reach and regional resilience.

Example:

A company runs its main Aurora database in Ohio and wants a secondary cluster in another region for failover and faster access for users in that region. After adding the region, Aurora keeps the secondary cluster synchronized with the primary cluster. If the primary region has a problem, the secondary region can be used to keep the application available.

Lab Steps

Step 1 – Open the Existing Aurora Cluster

- Sign in to the AWS Management Console.
- Open **Amazon RDS**.
- Select the existing **Amazon Aurora** cluster in the Ohio region.
- Open **Actions**.

Step 2 – Add an AWS Region

- Click **Add region**.

The screenshot shows the AWS Aurora and RDS console for a global database cluster named 'database-2'. The 'Actions' menu is open, and 'Add AWS Region' is highlighted. The console displays the primary region as 'us-east-1' with 2 instances. The configuration tab is selected, showing details like DB cluster role, engine version, and authentication settings.

- Choose the target AWS Region for the secondary cluster.
- Enter name for **Global Database Identifier**.
- Choose **Secondary Region**.

The screenshot shows the 'Add a region' page in the AWS Aurora and RDS console. The 'Global database identifier' is set to 'rds-serverless-global'. The 'Secondary region' is set to 'Asia Pacific (Mumbai)'. The 'Cluster storage configuration' section shows 'Aurora Standard' as the selected option.

- Review the cluster creation options for the new region.

Step 3 – Configure the Secondary Region

- Create or select a **VPC** in the target Region.
- Create or select a **database subnet group** in that Region.

Connectivity [Info](#)

Virtual private cloud (VPC) [Info](#)
Choose the VPC. The VPC defines the virtual networking environment for this DB cluster.

Default VPC (vpc-072c4acd761c5b942)
[→ Subnets: 2 Availability Zones](#)

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

ⓘ After a database is created, you can't change its VPC.

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB cluster can use in the VPC that you selected.

default
3 Subnets, 3 Availability Zones

Public access [Info](#)

☐ Yes
RDS assigns a public IP address to the cluster. Amazon EC2 instances and other resources outside of the VPC can connect to your cluster. Resources inside the VPC can also connect to the cluster. Choose one or more VPC security groups that specify which resources can connect to the cluster.

☒ No
RDS doesn't assign a public IP address to the cluster. Only Amazon EC2 instances and other resources inside the VPC can connect to your cluster. Choose one or more VPC security groups that specify which resources can connect to the cluster.

Existing VPC security groups
Choose one or more options

default [×](#)

Availability Zone [Info](#)

No preference

- Configure the new Aurora cluster settings.
- Set the **Aurora Capacity Units** if the cluster uses serverless.

Instance configuration
The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ Hide filters

☐ Include previous generation classes

☒ Aurora serverless
☐ Memory optimized classes (includes r classes)

Instance type

Aurora serverless
Instant scaling for even the most demanding workloads.

Capacity range [Info](#)
Database capacity is measured in Aurora Capacity Units (ACUs). 1 ACU provides 2 GiB of memory and corresponding compute and networking. The platform version has been updated to Version 4 offering scaling up to 256 ACUs, and performance improvement up to 30% compared to version 3.

Minimum capacity (ACUs)
0.5 (1 GiB)

Maximum capacity (ACUs)
1 (2 GiB)
1 to 128 in increments of 0.5

Pause after inactivity [Info](#)
Clusters with a minimum capacity setting of 0 ACUs can be paused during inactivity. Specify the amount of time the DB instance can be idle before pausing. For more information, see [Pause Aurora Serverless DB Instances](#).

00:05:00
Enter a time from 5 minutes to 24 hours

ⓘ **Pause DB after inactivity**
Clusters with a secondary region don't support pausing after inactivity when the minimum ACUs are set to 0.

- Keep the configuration aligned with the primary cluster where needed.

Step 4 – Create the Global Database Cluster

- Confirm the settings for the new Region.
- Create the additional Aurora cluster.

Backup

Backup retention period [Info](#)
The number of days (1-35) for which automatic backups are kept.
1 day

Backup tags [Info](#)
Copies tags from the source database to the automated backup and snapshots respectively.

☐ Copy tags to automated backup
This is a one-time setting. Future tag modifications need manual updates.

☒ Copy tags to snapshots

Maintenance

Auto minor version upgrade [Info](#)
☒ Enable auto minor version upgrade
Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see [Automatically upgrading the minor engine version documentation](#).

Maintenance window [Info](#)
Select the period you want pending modifications or maintenance applied to the database by Amazon RDS.

☐ Choose a window

☒ No preference

☐ Enable deletion protection
Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

[Cancel](#) [Add region](#)

- Wait for the secondary cluster to be provisioned.
- Verify that the new cluster appears in the selected Region.

Step 5 – Verify Synchronization

- Confirm that the primary cluster remains in the Ohio Region.
- Confirm that the secondary cluster is in the new Region.

DB Identifier	Status	Role	Engine	Region	Size	Recommendations	CPU
rds-serverless-global	Available	Global da...	Aurora My...	2 regions	2 clusters		-
database-2	Available	Primary cl...	Aurora My...	us-east-1	2 instances	2 Informational	-
database-2-instance-1	Available	Writer ins...	Aurora My...	us-east-1d	Aurora ser...		46.6
database-2-instance-1-reader	Available	Reader ins...	Aurora My...	us-east-1a	Aurora ser...		44.3
rds-serverless-global-cluster-1	Available	Secondar...	Aurora My...	ap-south-1	1 instance	2 Informational	-
rds-serverless-global-instance-1	Incompatible-n...	Reader ins...	Aurora My...	-	Aurora ser...		-

Global database

Configuration

Engine
Aurora MySQL

Engine version
8.0.mysql_aurora.3.12.0

RDS Extended Support
Disabled

Availability

Encryption
Enabled

AWS Regions
us-east-1 (N. Virginia)
ap-south-1 (Mumbai)

- Verify that changes from the primary region are synchronized to the secondary region.
- Confirm that the Aurora Global Database is now active.

Verification

- The existing Aurora cluster in Ohio was used as the primary cluster.
- A second AWS Region was added successfully.
- A secondary Aurora cluster was created in the new Region.

- The secondary Region used its own VPC and subnet group.
- Changes in the primary cluster were synchronized to the secondary cluster.
- The Aurora Global Database setup was completed successfully.